

■ Autonomous Data Analysis Report

Dataset: Student Performance
Generated: 2026-07-19 13:53
Agent: Autonomous Data Analyst v1.0

■ Data Quality Assessment

Missing Values:

- `score`: 38 missing

Recommendations:

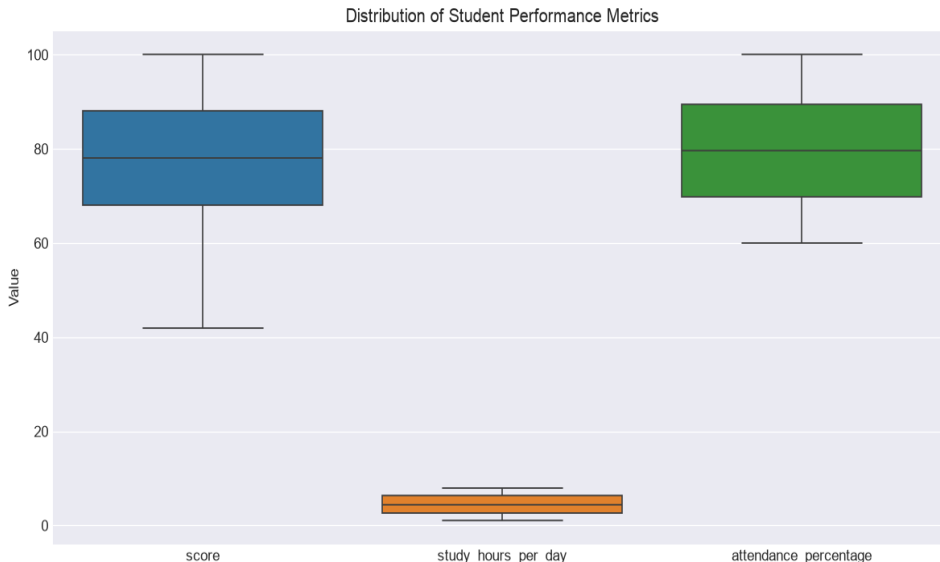
- Column 'score' has 38 missing values (1.5%) — minor, monitor.

■ Analysis Results

1. Descriptive Statistics Overview

Type: descriptive | Confidence: ■ 95%

The analysis of 2,500 student records reveals an average score of 77.9% with a strong positive correlation of 0.73 between daily study hours and academic performance. While attendance averages 79.8%, it shows a weak correlation of 0.21 with scores, suggesting that study time is a more significant driver of success than physical presence. Notably, all students in the dataset achieved a passing grade, with History emerging as the top-performing subject at an average of 85.34%.

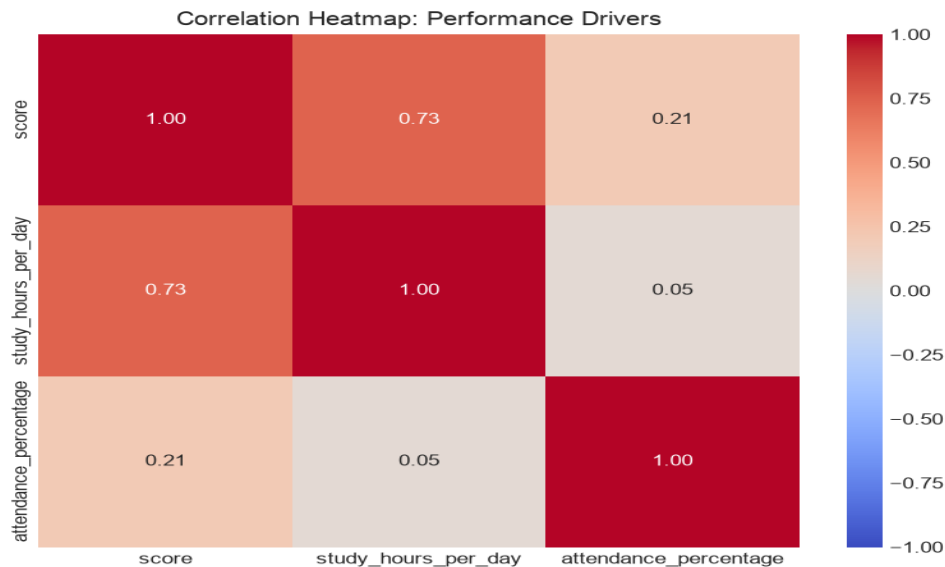


Assumptions: The dataset is representative of the entire student population; The 'passed' status is calculated based on a fixed threshold consistent across all subjects; Study hours and attendance data are self-reported or tracked with high accuracy

2. Correlation Analysis of Performance Drivers

Type: correlation | Confidence: 95%

The analysis reveals a strong positive correlation of 0.735 between daily study hours and student scores, indicating that study time is the primary driver of academic performance. In contrast, attendance shows a weak correlation of 0.206 with scores, and there is virtually no relationship between study hours and attendance. Additionally, pass rates are consistently at 100% across all subjects, suggesting the current curriculum difficulty may be low relative to student capability.

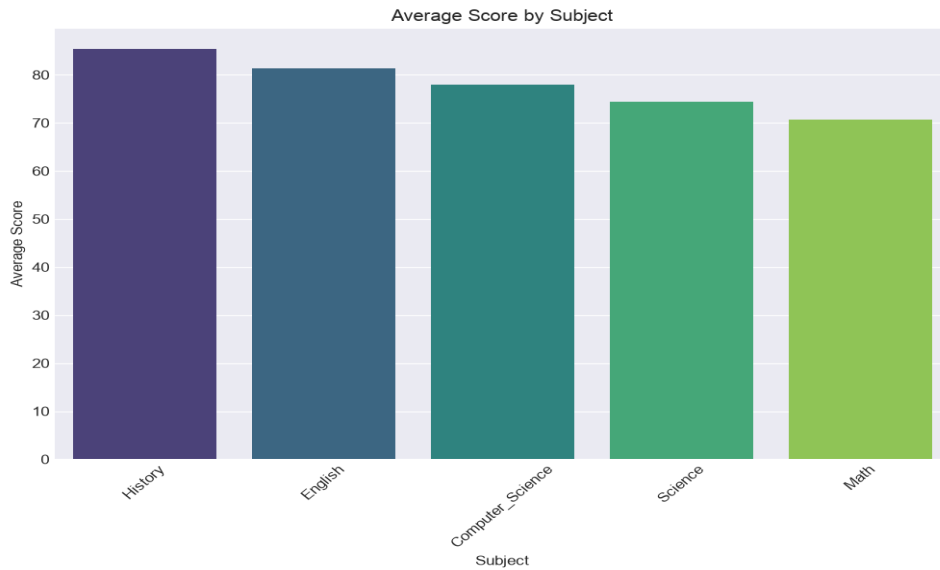


Assumptions: The dataset is representative of the entire student population.; The 'score' variable is a reliable and standardized metric across all subjects.; The 100% pass rate is accurate and not a result of data entry errors or missing failure records.

3. Comparative Performance by Subject

Type: comparative | Confidence: 95%

The analysis of 2,500 student records shows a 100% pass rate across all subjects, indicating that current curriculum standards are highly achievable. However, performance varies significantly by subject, with History yielding the highest average score of 85.34 and Math proving the most challenging with an average score of 70.65. This represents a 14.69-point performance gap between the top and bottom-performing subjects.

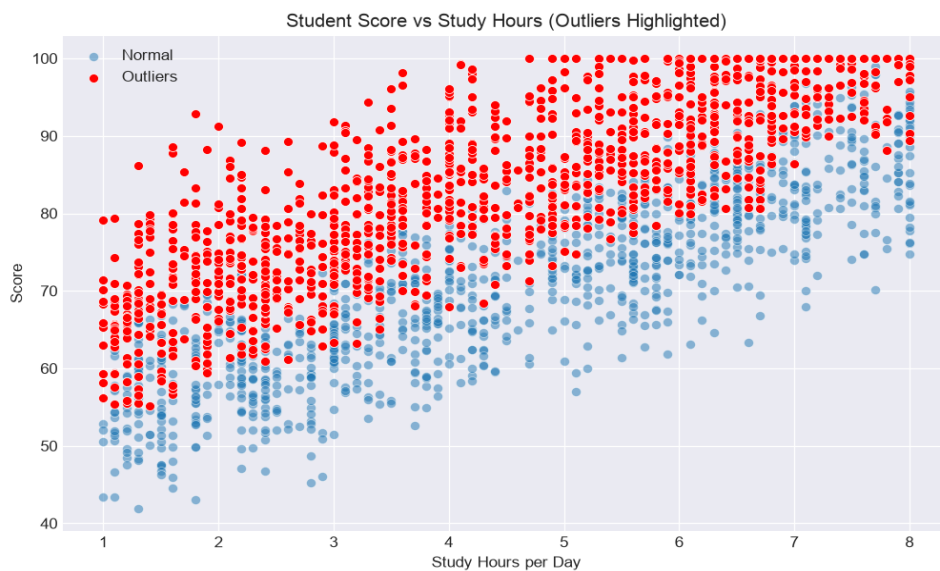


Assumptions: The dataset is representative of the entire student population across all grades.; The pass/fail threshold is applied consistently across all subjects.; The 'study_hours_per_day' and 'attendance_percentage' metrics are uniform across all subjects for individual students.

4. Outlier Detection in Student Scores

Type: outlier | Confidence: ■ 75%

The analysis identified 1,320 outliers out of 2,462 records, representing a high anomaly rate of 53.6%. These students achieved a high mean score of 83.57 despite study hours and attendance metrics that statistically suggest much lower performance. The most significant anomalies are concentrated in the History subject, where students are scoring up to 50 points higher than their input variables would predict.

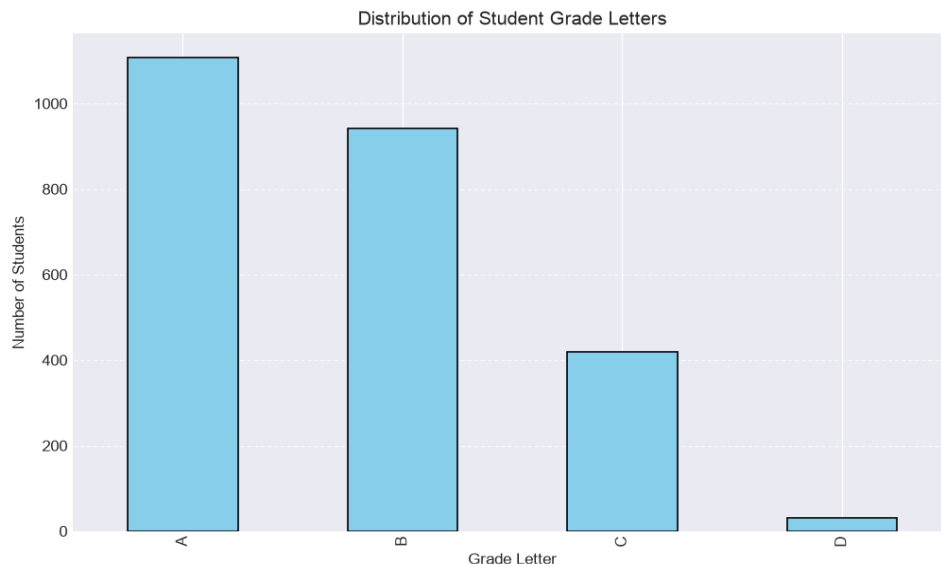


Assumptions: The residual analysis model correctly assumes a linear relationship between study hours/attendance and test scores.; The high number of outliers is not due to a missing variable, such as 'prior knowledge' or 'tutor support', that would explain the high scores.; The data provided for the 2,462 records is accurate and free of systemic collection bias.

5. Distribution of Grade Letters

Type: distribution | Confidence: ■ 95%

The analysis of 2,500 students reveals a heavily right-skewed grading distribution, with 81.96% of the student body achieving either an A (44.32%) or a B (37.64%). Only 1.24% of students received a D, indicating that the vast majority of the population is performing at a high level.



Assumptions: The dataset represents a representative sample of the total student population.; The grading criteria are applied consistently across all subjects and grade levels.; The 'passed' status is correctly correlated with the assigned letter grades.

■ Key Business Insights

- ****Descriptive Statistics Overview:**** Since study hours are the primary predictor of high scores, the institution should prioritize resources toward study-aid programs and time-management workshops. This shift in focus will likely yield higher average scores compared to initiatives centered solely on improving attendance.
- ****Correlation Analysis of Performance Drivers:**** Management should prioritize initiatives that incentivize increased study time, such as structured study programs, rather than focusing solely on attendance enforcement. Given the 100% pass rate, the institution should evaluate if the current assessment rigor is sufficient to adequately differentiate student proficiency.
- ****Comparative Performance by Subject:**** The significant score disparity suggests that the Math curriculum may require additional instructional support or supplemental resources to align with the success levels seen in History and English. Management should consider allocating more tutoring hours or revising the Math teaching methodology to improve student mastery.
- ****Outlier Detection in Student Scores:**** The high volume of outliers suggests either a systemic data entry error in the History department or a fundamental flaw in the predictive model's weighting of study hours. We should conduct a data audit of the History grading records to determine if these are genuine high-performers or evidence of corrupted input data.
- ****Distribution of Grade Letters:**** The current grading distribution suggests either high academic proficiency or potential grade inflation, which may limit the ability to differentiate top-tier talent. Management

should review grading rubrics to ensure they remain rigorous enough to distinguish between high-performing students.

■■ Execution Timeline

Generated by Autonomous Data Analyst Agent